01 Chemical Equilibrium and Reaction Rates Problem-solving Approach

Basis: `evidence/che/notes/EQ.txt` (EQ1–EQ10, item-by-item listing of question stems, key and exam reports), `tools/parttags_CHE.txt` (item tags), `tools/catalogue_CHE.json` (pattern definitions), `evidence/che/notes/INDEX.md` (mark allocation statistics). Resources available to students are taken from `CHE/papers/2025_CHE_DataBooklet.pdf`, with spot checks of the 2016, 2019, 2022 and 2024 data booklets. Exam question numbering format: `2024S-Q30(a)` = 2024 Section Two (short answer) Question 30(a); `2019E-Q37(e)` = Section Three (extended answer). `S` and `E` only indicate paper section, not Calculator-free or assumed. ★ markers: by INDEX mark ranking, approximately the top quarter are selected (top 3 of the 10 patterns in this topic): EQ4 (98 marks), EQ5 (58 marks), EQ1 (36 marks). Review status: independent review completed (against exam questions, key and exam reports), record at `CHE/methods/审校记录.md`. The key for 2019E-Q36(e) is a copyright placeholder in the PDF and could not be verified; noted in Pattern 10.

First, remember: where markers repeatedly deduct marks

- Explanations must state both the "direction" and the "reason": writing only shift to the right does not earn full marks; the key requires explaining why using collision theory or forward and reverse rates (for the difference between EQ4 and EQ5, see Pattern 4 and Pattern 5).
- When a question asks about rate, state whether it is the forward reaction rate or the reverse reaction rate: the key marks these separately (2017S-Q28, 2022S-Q26, 2024S-Q26).
- Only temperature changes K: concentration, pressure, volume and catalysts do not change K (the key for 2016E-Q41(b) is No change in K).
- Exam reports have very little specific comment on this topic: reports mostly review whole sections. The 2022 report said Question 26 (predicting the result of an equilibrium shift) was one of the best-answered questions in Section Two; the 2024 report listed Question 30 (explaining ocean acidification with equations) as one of the hardest questions in Section Two. Therefore, the "common errors" for each pattern below are based mainly on mark-deduction points in the key; report wording is quoted only where there is evidence.

What the data booklet provides (resources available to students in the exam)

- $K_w = 1.00 \times 10^{-14}$ is given only at 25 °C; K at other temperatures must be given in the question stem (e.g. 2017S-Q31(c) gives a table of K from 0 °C to 100 °C).
- The data booklet does not contain a table of K values (nor a Ka table); any K appearing in a question must be provided in the question stem.
- Colours table (Colours of selected substances): Co^{2+} pink, $CoCl_4^{2-}$ blue (given in the question stem), Fe^{3+} pale brown, Fe^{2+} pale green, Cu^{2+} blue, CrO_4^{2-} yellow, $Cr_2O_7^{2-}$ orange, MnO_4^- purple; coloured halogens are listed only as elements (Cl_2(g) yellow-green, Br_2(ℓ) red, I_2(s) grey, I_2(g) purple); for orange or brown aqueous solutions, follow the key (e.g. 2023S-Q30(c), 2024S-Q27(b)).
- Solubility rules and the standard electrode potential table are less relevant to this topic; among the EQ items, only EQ9 directly relies on the colours table.

Pattern 1: Explaining reaction rates using collision theory ★

Recognise: "Explain the effect of … on the rate … by applying collision theory" "Use collision theory to explain …" "Support your explanation with a labelled diagram". Common contexts: temperature (Boltzmann diagram), concentration, surface area of reactant powder (2022E-Q35(b)), catalyst (2024E-Q37(b)), decrease in reactant during a reaction (2019S-Q34(b)).

Steps

1. State the conclusion first: the rate increases or decreases, and which rate it is (forward reaction or overall reaction).
2. Match one causal chain to the factor:

- Increased concentration or pressure: more particles per unit volume, so collision frequency increases.
- Increased temperature: average kinetic energy increases, so the proportion of particles with energy ≥ E_a increases (the proportion of effective collisions increases); at the same time, particles move faster, so collision frequency increases.
- Catalyst: provides a reaction pathway with lower activation energy E_a, so the proportion of collisions with energy ≥ E_a increases.
- Increased surface area: the area of the reactant available for contact increases, so collision frequency increases.

3. For temperature questions, you must write two reasons (proportion of effective collisions + collision frequency), not just "more collisions".
4. When a diagram is required, mark E_a and the curve after the temperature change on a Boltzmann distribution diagram (after heating, the curve shifts right and the peak becomes lower and broader), and label the axes.
5. When explaining the shape of a graph, link the reason to the slope or plateau: high concentration gives a steep initial slope; as reactant is consumed the rate decreases, and the curve approaches a plateau.

Mark-earning points (typical wording in the key): ✓ average kinetic energy / proportion of particles with energy above E_a (first point in 2018E-Q37(a)) ✓ increased proportion of effective collisions (second and third points) ✓ increased collision frequency, therefore increased overall rate (fourth and fifth points) ✓ clearly labelled Boltzmann diagram (the diagram in 2018E-Q37(a) is worth 2 marks) ✓ 2018E-Q37(c): increased concentration → more particles per unit volume → increased collision frequency → increased rate, steeper slope, and explain the plateau by the reaction finishing faster. ✓ 2019S-Q34(b) (6 marks): rate is represented by the rate of change in CO_2 volume; decreasing concentration and decreasing surface area both lower collision frequency; after 4 minutes the reactant is used up and the curve plateaus.

Common errors (exam reports): Exam reports do not comment on each collision theory question individually. The 2021 report said the whole of Question 37 (18 marks, average 39.5%) could discriminate between candidates and that the calculations required non-standard methods; that average is for the whole question, not for (aii) (3 marks) alone. The following common errors are summarised from the key:

- Writing only "more collisions" without stating that they are effective collisions (energy ≥ E_a).
- For temperature questions, writing only collision frequency and omitting average kinetic energy and energy distribution.
- For catalyst questions, writing only "faster" without stating that E_a is lowered.
- Boltzmann diagram without E_a marked, or with the curve direction drawn incorrectly.

Exam questions: 2016E-Q39(c), 2018E-Q37(a), 2018E-Q37(c), 2019S-Q34(b), 2020S-Q34(b), 2021E-Q37(a)(ii), 2022E-Q35(b), 2024E-Q37(b), 2025E-Q38(b)

Pattern 2: Energy profile diagrams

Recognise: "sketch an energy profile diagram" "label the activation energy and change in enthalpy" "indicate by a dashed line the change if an inhibitor ..." . You may also be asked to read ΔH and E_a from a given diagram, or decide whether a reaction is endothermic or exothermic.

Steps

1. Draw the axes: label the vertical axis Energy / Enthalpy (H) (with kJ units; the key for 2021S-Q32(b) requires the y-axis unit to be written), and the horizontal axis Reaction pathway / Reaction progress.
2. The reactant horizontal line is on the left, the product horizontal line on the right. For an exothermic reaction the products are lower than the reactants (ΔH is negative); for an endothermic reaction it is the opposite.
3. The top of the peak is the activated complex (transition state). E_a = peak energy − reactant energy;

mark it on the diagram.

4. Mark ΔH (the energy difference between reactants and products); note the magnitude: if E_a is much smaller than |ΔH|, draw it to scale (e.g. the explosive-type reaction in 2017S-Q33(a), and the reaction between "pool chlorine" and "pool acid" in 2019S-Q27(c), where the key requires E_a to be very small relative to ΔH).
5. Catalyst / inhibitor: a catalyst lowers the peak (reduces E_a); an inhibitor raises the peak, with ΔH unchanged (2017S-Q33(b)).

Mark-earning points (typical wording in the key): ✓ axes and labels (2019S-Q27(c), 2021S-Q32(b)) ✓ correct positions of reactants and products (2017S-Q33(a): product energy lower than reactant energy) ✓ both E_a and ΔH marked (2022E-Q35(a): E_a and ΔH correctly labelled) ✓ correct magnitude and sign: peak about 600 kJ and ΔH about −92 kJ in 2021S-Q32(b); 2019S-Q27(c) requires ΔH to be negative and relatively large, with E_a very small ✓ inhibitor question (2017S-Q33(b)): dashed peak is higher, reactant and product energies unchanged

Common errors (exam reports): The evidence contains no specific report comment on energy profile diagrams. Points where the key repeatedly deducts marks:

- Missing axis labels or missing y-axis units (2021S-Q32(b)).
- Marking only E_a and not ΔH, or omitting the negative sign on ΔH.
- Drawing endothermic and exothermic the wrong way round (products higher than reactants but labelled exothermic).
- The key for 2022E-Q35(a) requires the magnitudes of E_a and ΔH to be reasonable; E_a must not be drawn at about the same height as the reactants.

Exam questions: 2016E-Q41(f), 2017S-Q33(a), 2017S-Q33(b), 2019S-Q27(c), 2021S-Q32(b), 2022E-Q35(a)

Pattern 3: Characteristics of dynamic equilibrium and open/closed systems

Recognition: "State / describe the characteristics of a dynamic equilibrium", "State three inferences …", often accompanied by isotope labelling (e.g. ^{15}N tracing) or colour and concentration data.

Steps

1. Dynamic equilibrium requires two points: macroscopic properties (colour, concentration, pressure) no longer change with time; at the microscopic level, the forward and reverse reactions are still occurring and their rates are equal.
2. Isotope tracing evidence: labelled atoms appear in both reactants and products → the forward and reverse reactions are both occurring; the labelled ratio remains constant → a new equilibrium has been established and the rates are equal (2025S-Q30(a)).
3. Dynamic equilibrium does not require the concentrations of all species to be equal, only the forward and reverse rates to be equal (catalogue definition: concentrations are not necessarily equal).
4. If the question asks about open/closed systems: an open system exchanges matter with the surroundings, while a closed system does not exchange matter; in the evidence, this point appears only in the catalogue definition, with no corresponding key, so when answering write only the definitional explanation.

Marking points (typical key wording): ✓ 2025S-Q30(a) (3 marks): One, both the forward and reverse reactions are occurring (dynamic equilibrium); two, some of the injected ^{15}N nitrogen gas is converted to ammonia; three, the constant ratio shows that a new equilibrium has been established, with the forward and reverse rates equal. The key notes: ^{15}N must be explicitly stated, otherwise full marks cannot be awarded. ✓ 2018E-Q38(c)(i) (1 mark): The change at T_1 can be any one of "the system has reached equilibrium", addition of a catalyst, or "no change" (the key accepts multiple answers).

Common errors (Examiner reports): The Examiner reports do not have specific comments on this pattern. Points repeatedly required by the key:

- Describing dynamic equilibrium as "the reaction has stopped" or "the concentrations are equal".

- The three inferences repeat each other (e.g. two statements both say "the forward and reverse reactions occur at the same time"), in which case only one mark can be awarded.

Past exam questions: 2018E-Q38(c)(i), 2025S-Q30(a)

Pattern 4: Using Le Châtelier's Principle to predict equilibrium shifts ★

Recognition: "Predict the effect of ..." , "Use Le Châtelier's Principle to predict ..." , "circle increase / decrease / no effect" , and disturbances given in table form, requiring prediction of shift, rate, colour (2022S-Q26, 2024S-Q26).

Steps

1. Write the equilibrium equation, mark the sign of ΔH, states (solids and pure liquids are not included), and the gas coefficients on both sides.
2. Determine the type of disturbance:
    - Adding or removing a substance: shifts in the direction that counteracts the change (removing product → shifts right).
    - Changing volume (total pressure): look at the gas coefficients on both sides; decreasing volume → shifts to the side with fewer gas molecules (2025S-Q30(b)); increasing volume is the opposite (in 2018E-Q38(b), increasing volume decreases methane yield).
    - Changing temperature: the endothermic direction is promoted by heating (2017S-Q28(b), 2020S-Q34(a)).
    - Adding a catalyst: does not shift the equilibrium, does not change the yield, but increases the forward and reverse rates at the same time (key for 2018E-Q38(b) is no effect).
    - Adding an inert gas: at constant volume it does not change the equilibrium; at constant pressure, the volume change must be considered (there is no corresponding question in the evidence, so state the premise when answering).
3. Write the direction of shift and the change in the amount or concentration of the target species; when required by the question, write the rate (whether the forward or reverse reaction rate changes relative to the original).
4. Colour questions: using the colour table in the data booklet, state whether the colour becomes deeper or paler, and link it to the concentration change of that species (key for 2022S-Q26: adding a small amount of $FeCl_3$ shifts the equilibrium right and the solution becomes a deeper red; heating shifts the equilibrium left and the solution becomes more brown or orange; adding water and adding concentrated Na_3PO_4 shift the equilibrium left and the solution is a very pale brown).

Marking points (typical key wording): ✓ 2022S-Q26 (12 marks): marks are awarded separately for three columns (direction of shift, forward reaction rate, colour); each column corresponds to four changes and is worth 4 marks. ✓ 2024S-Q26 (9 marks): three columns worth 3 marks each, corresponding to three changes. ✓ 2018E-Q38(b) (5 marks): catalyst has no effect; increasing volume → methane yield decreases; heating → decreases; removing methane → increases; lowering CO_2 partial pressure → decreases. ✓ 2016E-Q39(c) (2 marks): when pressure is increased, rate increases and yield increases. ✓ 2017S-Q28(a) (6 marks): after adding hydrochloric acid, pH decreases and the forward reaction rate increases; after adding $Pb(NO_3)_2$, both decrease; after adding water, rate decreases and pH increases. ✓ 2025S-Q30(b) (4 marks): decreasing volume causes the amount of ^{15}N in nitrogen gas to decrease and the amount in ammonia to increase, and states that the equilibrium shifts to the side with fewer gas molecules.

Common errors (Examiner reports): The 2022 report states that Question 26 (predicting the outcome of equilibrium changes) was one of the best-answered questions in Section Two, indicating that the routine steps for this type of question were well understood. The following common errors are summarised from the key:

- Writing only the shift and not the rate (2022S-Q26, 2024S-Q26 are marked by column).
- Writing that a catalyst "shifts the equilibrium to the right" .

- Mixing up the effect of temperature with the change in K (see Pattern 8).
- In volume-change questions, using "concentration increases" instead of "pressure increases", ignoring the gas coefficients.

Past exam questions: 2016E-Q39(c), 2016E-Q41(e), 2017S-Q28(a), 2017S-Q28(b), 2018E-Q38(b), 2018E-Q38(c)(iii), 2018S-Q32, 2019E-Q36(d), 2019E-Q36(f), 2019S-Q31(c), 2020E-Q40(d), 2020S-Q29(b), 2020S-Q29(c), 2020S-Q31(c), 2020S-Q34(a), 2021S-Q29(a), 2021S-Q32(d), 2022S-Q26, 2023S-Q28(b), 2024S-Q26, 2024S-Q30(a), 2024S-Q30(b), 2025S-Q30(b)

Pattern 5: Using collision theory/forward and reverse rates to explain equilibrium shifts ★

Recognition: "Use collision theory to explain the change in concentrations …", "Explain why the equilibrium shifts …", "Justify your answer using collision theory". The question stem usually first gives a disturbance (removing a substance, adding OH^-, decreasing volume).

Steps

1. State the disturbance and identify which rate changes first, because of the change in collision frequency.
2. State the relative size of the forward and reverse reaction rates ("the rate of the reverse reaction is greater than the forward reaction"), and explain that the other rate is initially unaffected.
3. State the direction in which the equilibrium shifts until the forward and reverse rates are equal again, forming a new equilibrium.
4. Explain the change in concentration of the relevant species at the new equilibrium (increased or decreased relative to the original equilibrium).

Example structure (compiled from the key for 2020S-Q29(b) and 2021S-Q32(c)): remove X → collision frequency of X decreases → the rate related to X decreases → the rate in the other direction is greater → the equilibrium shifts in the direction that produces X until the rates are equal.

Marking points (typical key wording): ✓ 2020S-Q29(b) (5 marks): CH_4 decreases → collision frequency decreases → forward reaction rate decreases → reverse reaction rate is greater → equilibrium shifts left. ✓ 2021S-Q32(c) (5 marks): at the instant NH_3 is removed → reverse reaction rate decreases → forward reaction rate is faster → NH_3 increases; H_2 and N_2 decrease → new equilibrium. ✓ 2025S-Q29(b) (5 marks): write the equation containing OH^- (e.g. $CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$); OH^- increases; the forward reaction rate decreases (or the reverse reaction rate increases); at the new equilibrium $[H_3O^+]$ decreases. ✓ 2025S-Q32(c) (4 marks): the equilibrium equation in the question stem is $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$, and the forward reaction refers to CrO_4^{2-} forming $Cr_2O_7^{2-}$. OH^- lowers the H^+ concentration → collision frequency decreases → the forward reaction rate decreases relative to the reverse reaction → at the new equilibrium $[CrO_4^{2-}]$ increases and $[Cr_2O_7^{2-}]$ decreases.

Common errors (Examiner reports): The Examiner reports do not have specific comments on this pattern. Points repeatedly penalised by the key:

- Writing only "the rate decreases" without specifying whether it is the forward or reverse reaction.
- Omitting the step "the other rate is initially unchanged", so the cause-and-effect chain is incomplete.
- Writing "the equilibrium shifts" but not writing collision theory or a comparison of the forward and reverse rates.
- 2025S-Q29(b) requires writing the equilibrium equation containing OH^-; without the equation, full marks cannot be awarded.

Past exam questions: 2016E-Q39(d), 2017S-Q28(a), 2017S-Q28(b), 2019E-Q36(g), 2020S-Q29(b), 2021S-Q32(c), 2022S-Q26, 2024S-Q26, 2025S-Q29(b), 2025S-Q32(c)

Pattern 6: Concentration/rate–time graphs

Identify: "On the axes below, sketch the change in concentration ..." "draw the forward and reverse rates ..." "show on the graph the effect of ..." . The disturbance may be temperature, volume, adding or removing a substance.

Steps

1. First draw the equilibrium section before the disturbance: the forward and reverse rates are equal, and concentration is a horizontal line.
2. Determine the type of change:
    - Changing volume or adding/removing a substance: concentration or rate changes discontinuously (a sudden change) at the instant of disturbance.
    - Concentration of a substance during the reaction: changes gradually from the moment of disturbance and cannot jump.
    - Increase in temperature: the initial slope of the rate becomes steeper, and the curve reaches the plateau earlier (2019S-Q34(c); this question is the change in CO_2 volume over time, not reversible equilibrium; the key only looks at changes in slope and plateau).
3. Stoichiometric ratio determines magnitude: draw the ratio of changes according to the coefficients in the equation (in 2016E-Q41(d), the ratio of change for O_2 to N_2 is 2:1; in 2025S-Q32(a), the rate of decrease of $Cr_2O_7^{2-}$ is about half the rate of increase of CrO_4^{2-}).
4. The horizontal section of the new equilibrium starts at the same time (the key for 2025S-Q32(a) requires the two curves to stabilise at the same time).
5. In rate graphs, the forward and reverse rate curves intersect at the equilibrium point; after the disturbance the two lines separate, intersect again, and extend horizontally (2016E-Q41(ci)).

Marking points (typical key wording): ✓ 2016E-Q41(ci) (3 marks): correct shape and direction of the forward and reverse curves; the two lines intersect at point A; A to B is a straight line. ✓ 2016E-Q41(d) (5 marks): correct curve shapes for $N_2(g)$ and $O_2(g)$; $NO_2(g)$ starts from zero; ratio 2:1; E_1 to E_2 is a horizontal line. ✓ 2025S-Q32(a) (4 marks): correct curve shapes for $Cr_2O_7^{2-}$ and CrO_4^{2-}; the two curves reach the new equilibrium at the same time; the rate of decrease of $Cr_2O_7^{2-}$ is about half the rate of increase of CrO_4^{2-}. ✓ 2018E-Q38(ciii) (3 marks): after cooling, CO_2 and H_2 decrease, CH_4 and H_2O increase, and a new equilibrium is reached; the key requires the changes to be labelled proportionally (the change in H_2 is about 4 times that of CO_2, and the change in H_2O is about 2 times that of CH_4).

Common errors (exam report): The exam report has no specific comments on this pattern. Points repeatedly penalised in the key:

- For temperature questions, only drawing the sudden change in concentration and forgetting to draw the change in rate or slope.
- Incorrect stoichiometric ratio (e.g. the same change in O_2 and N_2).
- The curve for a product (initial zero) does not start from zero.
- The two curves do not reach the new equilibrium at the same time.

Past questions: 2016E-Q41(c)(i), 2016E-Q41(c)(ii), 2016E-Q41(d), 2016E-Q41(e), 2018E-Q37(b), 2018E-Q38(c)(i), 2018E-Q38(c)(ii), 2018E-Q38(c)(iii), 2019S-Q34(c), 2021S-Q32(d), 2025S-Q32(a)

Pattern 7: Equilibrium constant expressions

Identify: "Write the equilibrium expression for ..." "Write the equilibrium constant expression for ..." "K = ..." ; it may also require working backwards from the K expression to the equilibrium equation.

Steps

1. First write the balanced equilibrium equation, indicating states (aq for aqueous solution, g for gas, ℓ for liquid).
2. K = (product of product concentrations raised to powers) ÷ (product of reactant concentrations raised to powers), where the powers equal the stoichiometric coefficients.

3. Do not include solids and pure liquids: self-ionisation of water is written as $K = [H_3O^+][OH^-]$ (2017S-Q31(b)); water itself does not appear.
4. When gases use partial pressure or concentration, the question or key may allow both forms, but be consistent within the same question.
5. Write the K = symbol and keep brackets and exponents clear.

Marking points (typical key wording): ✓ 2016E-Q41(a) (2 marks): correct structure (products ÷ reactants) and correct exponents; one incorrect or missing exponent loses 1 mark (key states "one superscript incorrect or missing"). ✓ 2019E-Q36(b) (2 marks): 1 mark for writing "K =", 1 mark for correct structure of the concentration expression. ✓ 2020S-Q29(a) (2 marks): one minor error loses 1 mark (key states "one minor error is 1 mark only"). ✓ 2023S-Q34(a) (2 marks): marks awarded for the structure $K = [CH_3(CH)_4COO^-][H_3O^+]/[CH_3(CH)_4COOH]$.

Common errors (exam report): The exam report has no specific comments on this pattern. Points repeatedly penalised in the key:

- Omitting K =.
- Missing or incorrect exponents (one occurrence loses 1 mark).
- Writing water or a pure solid into the expression (the key for 2017S-Q31(b) accepts only $[H_3O^+][OH^-]$ or $[H^+][OH^-]$).
- Writing pH or pKa values into the expression.

Past questions: 2016E-Q41(a), 2017S-Q31(b), 2018S-Q31(a), 2019E-Q36(b), 2020S-Q29(a), 2021S-Q32(a), 2023S-Q34(a)

Pattern 8: Magnitude of K and the effect of temperature on K

Identify: "What does the value of K indicate about ..." "predict the effect of an increase in temperature on the value of K" "what happens to K after the pressure is lowered".

Steps

1. Large K → products dominate, equilibrium position lies to the right; small K → reactants dominate (key for 2016E-Q41(f): K is small, reactants dominate, activation energy is large).
2. Effect of temperature change on K: first use the sign of ΔH and Le Châtelier to determine the direction of shift, then determine whether K increases or decreases. For an endothermic reaction, increasing the temperature increases K (e.g. self-ionisation of water; in the table for 2017S-Q31(c), K increases as the temperature increases); for an exothermic reaction, increasing the temperature decreases K (2020S-Q29(c)).
3. Changes in concentration, pressure, volume or catalyst shift the equilibrium but do not change K (2016E-Q41(b): after pressure is reduced, K is unchanged).
4. When explaining, write "temperature → direction of shift → change in amount of products or reactants → change in the value of K".

Marking points (typical key wording): ✓ 2020S-Q29(c) (4 marks): K decreases (1 mark); from the graph, the amount of H_2 decreases when the temperature is increased (1 mark); any two of the following: $[H_2S]$ and $[CH_4]$ increase; $[H_2]$ decreases; the reverse reaction is favoured (2 marks). ✓ 2016E-Q41(b) (1 mark): No change in K. ✓ 2017S-Q31(c) (1 mark): Kw is very small, indicating that only a very small amount of product exists at equilibrium and the system mainly exists as molecular water (key accepts other reasonable answers).

Common errors (exam report): The exam report has no specific comments on this pattern. Points repeatedly penalised in the key:

- Attributing a change in K to concentration or pressure (only temperature changes K).
- Only saying "shift to the left" and not writing the change in the value of K.
- Writing "K increases" incorrectly as "reactants increase".

Past questions: 2016E-Q41(b), 2016E-Q41(f), 2017S-Q31(c), 2020S-Q29(c)

Pattern 9: Observable phenomena in equilibrium systems

Identify: "describe the change in colour ..." "predict the colour of the reaction mixture" "describe the change in colour that will occur in the solution as it shifts to the new equilibrium position" ; the question stem gives coloured ions or complexes.

Steps

1. First write the colour before the disturbance (using the question stem and the colour table in the data booklet): e.g. Fe^{3+} pale brown, $FeSCN^{2+}$ deep red (2022S-Q26); $CoCl_4^{2-}$ blue, $[Co(H_2O)_6]^{2+}$ pink (2024S-Q26); CrO_4^{2-} yellow, $Cr_2O_7^{2-}$ orange (2025S-Q32).
2. After determining the direction of equilibrium shift, determine whether the concentration of the coloured species increases or decreases, then write that the colour becomes darker, lighter, or changes to another colour.
3. Complete wording: initial colour → final colour (or darker/lighter), and state whether it is a solution or a solid.

Marking points (typical key wording): ✓ 2025S-Q32(b) (2 marks): initially orange-yellow (1 mark); as it shifts to the new equilibrium, the solution becomes more yellow (1 mark). ✓ 2024S-Q26 (9 marks): colour column is worth 3 marks in total, 1 mark for each change (three changes). ✓ 2022S-Q26 (12 marks): colour column is worth 4 marks in total, 1 mark for each change (four changes).

Common errors (exam report): The exam report has no specific comments on this pattern. Points repeatedly penalised in the key:

- Only writing "changes colour" and not writing the initial and final colours.
- Colour names inconsistent with the data booklet (e.g. writing the orange of $Cr_2O_7^{2-}$ as red).
- For the Fe^{3+} and SCN^- system, incorrectly relating the deepening of colour after "$FeCl_3$ is added" to the direction of equilibrium shift.

Past questions: 2022S-Q26, 2024S-Q26, 2025S-Q32(b)

Pattern 10: Equilibrium in Real-World Contexts

Recognise: "explain what will happen in the blood when ..." "use the equations to predict what might happen to ocean pH ..." "Use Le Châtelier's Principle and the sequence of equations ..." . Common contexts: ocean acidification (CO_2 dissolution → H_2CO_3 → HCO_3^- → $CaCO_3$ dissolution), blood buffering (CO_2 and HCO_3^-), lung respiration and pH.

Steps

1. Reason through each equation in the order given in the question: external change → shift in the first equilibrium → change in the intermediate species → shift in the next equilibrium → final change in $[H_3O^+]$ or pH.
2. For each step, state whether it shifts left or right, and give a reason using Le Châtelier or collision theory.
3. Relate the conclusion back to the context: pH decreases or increases; $CaCO_3$ formation increases or decreases.
4. If asked to write two consequences, the two consequences must be different and reasonable (the 2020S-Q31(b) key only required two different reasonable consequences).
5. Watch for secondary effects: as the 2024S-Q30(a) key notes, an increase in HCO_3^- may shift Equation 3 left, but $[H_3O^+]$ still increases overall, so the conclusion does not change.

Marking points (typical key wording): ✓ 2018S-Q32 (4 marks): $CO_2(aq)$ increases → Equation 2 shifts right → $[CO_3^{2-}]$ decreases → Equation 3 shifts left → $CaCO_3$ formation decreases. ✓ 2023S-Q28(b) (5 marks): CO_2 increases → H_2CO_3 increases (write the equation) → buffer equilibrium shifts right → $[H_3O^+]$ increases → blood pH decreases slightly. ✓ 2024S-Q30(a) (4 marks): CO_2 dissolution increases → Equation 1 shifts right; H_2CO_3 increases → Equation 2 shifts right; HCO_3^- increases → Equation 3 shifts right; $[H^+]$ increases, pH

decreases. ✓ 2024S-Q30(b) (2 marks): $[H^+]$ increases → Equation 4 shifts right → $CaCO_3$ is consumed, so the available amount decreases. ✓ 2020S-Q31(c) (4 marks): If the Kyoto Protocol is abandoned → CO_2 emissions increase → dissolution increases → the forward reaction is promoted → $[H_3O^+]$ increases → ocean pH decreases further.

Common errors (examiner reports): The 2023 report listed "defining and explaining a buffer system" as one of the hardest questions on that paper (2023S-Q28 in the same year is related to the blood buffering in this pattern; see Pattern 4 and AB4). The 2024 report listed Question 30 (explaining ocean acidification using the given equations) as one of the hardest questions in Section Two. The following common errors are summarised from the key:

- Writing only "CO_2 increases so pH decreases" without writing the equilibrium equations (the keys for 2023S-Q28(b) and 2024S-Q30(a) both require equations).
- Missing the direction of a shift in one step, causing a break in the conclusion.
- Ignoring the secondary effect in 2024S-Q30(a) that "an increase in HCO_3^- shifts Equation 3 left".
- 2019E-Q36(f) must be answered using the equilibrium shift principle; (g) must be explained using collision theory and rates, not only the equilibrium shift principle (2019 report). The key for 2019E-Q36(e)–(g) is a copyright placeholder in the PDF; the two concentration conditions required by (e) could not be verified, so answer according to the question stem and the steps in this pattern.

Past exam questions: 2018S-Q32, 2019E-Q36(f), 2019E-Q36(g), 2020S-Q31(b), 2020S-Q31(c), 2022E-Q38(c), 2023S-Q28(b), 2024S-Q30(a), 2024S-Q30(b)